

3.11 Solution: (a) Take the Bessel differential equation written in its standard self-adjoint form

$$\frac{1}{\rho} \frac{d}{d\rho} \left[\rho \frac{dJ_\nu(k\rho)}{d\rho} \right] + \left(k^2 - \frac{\nu^2}{\rho^2} \right) J_\nu(k\rho) = 0. \quad (1)$$

Multiplying both sides by $\rho J_\nu(k'\rho)$ and integrating with respect to ρ from 0 to a , we obtain

$$\int_0^a J_\nu(k'\rho) \frac{d}{d\rho} \left[\rho \frac{dJ_\nu(k\rho)}{d\rho} \right] d\rho + \int_0^a \left(k^2 - \frac{\nu^2}{\rho^2} \right) \rho J_\nu(k'\rho) J_\nu(k\rho) d\rho = 0. \quad (2)$$

Performing integration by parts on the first term, the relation becomes

$$\rho J_\nu(k'\rho) \frac{dJ_\nu(k\rho)}{d\rho} \Big|_0^a - \int_0^a \rho \frac{dJ_\nu(k'\rho)}{d\rho} \frac{dJ_\nu(k\rho)}{d\rho} d\rho + \int_0^a \left(k^2 - \frac{\nu^2}{\rho^2} \right) \rho J_\nu(k'\rho) J_\nu(k\rho) d\rho = 0. \quad (3)$$

Using the first boundary condition at the origin, the lower limit of the boundary term vanishes, $\rho J_\nu(k'\rho) \frac{dJ_\nu(k\rho)}{d\rho} \Big|_{\rho=0} = 0$. For the boundary term at the upper limit $\rho = a$, we have

$$\frac{dJ_\nu(k\rho)}{d\rho} \Big|_{\rho=a} = -\frac{\lambda}{a} J_\nu(ka). \quad (4)$$

Substituting this boundary condition into the integrated term yields the value $-\lambda J_\nu(k'a) J_\nu(ka)$. Thus, equation (3) can be written as

$$-\lambda J_\nu(k'a) J_\nu(ka) - \int_0^a \rho \frac{dJ_\nu(k'\rho)}{d\rho} \frac{dJ_\nu(k\rho)}{d\rho} d\rho + \int_0^a \left(k^2 - \frac{\nu^2}{\rho^2} \right) \rho J_\nu(k'\rho) J_\nu(k\rho) d\rho = 0. \quad (5)$$

By exchanging the parameters k and k' , we generate a symmetric relation

$$-\lambda J_\nu(ka) J_\nu(k'a) - \int_0^a \rho \frac{dJ_\nu(k\rho)}{d\rho} \frac{dJ_\nu(k'\rho)}{d\rho} d\rho + \int_0^a \left(k'^2 - \frac{\nu^2}{\rho^2} \right) \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho = 0. \quad (6)$$

Taking the difference between equation (5) and equation (6), the boundary terms and the first integral term identically cancel out, leaving

$$(k^2 - k'^2) \int_0^a \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho = 0. \quad (7)$$

This implies that for distinct eigenvalues ($k \neq k'$), the orthogonality condition must hold true,

$$\int_0^a \rho J_\nu(k\rho) J_\nu(k'\rho) d\rho = 0. \quad (8)$$

(b) Alternatively, the Bessel equation can be expanded in the following form, where primes denote differentiation with respect to ρ ,

$$\rho[\rho J'_\nu(k\rho)]' + (k^2 \rho^2 - \nu^2) J_\nu(k\rho) = 0. \quad (9)$$

Multiplying equation (9) by $J'_\nu(k\rho)$ and integrating over the domain $[0, a]$ gives

$$\int_0^a \rho J'_\nu(k\rho) [\rho J'_\nu(k\rho)]' d\rho + \int_0^a (k^2 \rho^2 - \nu^2) J'_\nu(k\rho) J_\nu(k\rho) d\rho = 0. \quad (10)$$

The first term can be integrated explicitly as a perfect differential[cite: 32]:

$$\frac{1}{2} [\rho J'_\nu(k\rho)]^2 \Big|_0^a = \frac{1}{2} a^2 \left(\frac{dJ_\nu(k\rho)}{d\rho} \right)^2 \Big|_{\rho=a} = \frac{1}{2} k^2 a^2 [J'_\nu(ka)]^2.$$

The second term in equation (10) can be computed using integration by parts,

$$\begin{aligned} \int_0^a (k^2 \rho^2 - \nu^2) J'_\nu(k\rho) J_\nu(k\rho) d\rho &= \frac{1}{2} \int_0^a (k^2 \rho^2 - \nu^2) d[J_\nu(k\rho)^2] \\ &= \frac{1}{2} (k^2 a^2 - \nu^2) J_\nu(ka)^2 - k^2 \int_0^a \rho J_\nu(k\rho)^2 d\rho. \end{aligned}$$

Isolating the normalization integral from the total combined expression leads to

$$\int_0^a \rho J_\nu(k\rho)^2 d\rho = \frac{a^2}{2} [J'_\nu(ka)]^2 + \frac{a^2}{2} \left(1 - \frac{\nu^2}{k^2 a^2} \right) J_\nu(ka)^2 \quad (11)$$

Using the notation $y_{\nu n} = ka$ for the discrete roots corresponding to eigenvalues, this becomes

$$\int_0^a \rho J_\nu \left(\frac{y_{\nu n}}{a} \rho \right)^2 d\rho = \frac{a^2}{2} \left[\left(1 - \frac{\nu^2}{y_{\nu n}^2} \right) J_\nu(y_{\nu n})^2 + \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2 \right]. \quad (12)$$

For a generalized Fourier-Bessel series expansion $f(\rho) = \sum_{n=1}^{\infty} A_n J_\nu \left(\frac{y_{\nu n}}{a} \rho \right)$, the coefficients A_n are determined by taking inner products with respect to the orthogonal weight $\rho J_\nu \left(\frac{y_{\nu n}}{a} \rho \right)$ across $[0, a]$. Using the boundary condition, we can relate the derivative value directly to the functional value at the boundary

$$\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} = -\frac{\lambda}{y_{\nu n}} J_\nu(y_{\nu n}) \quad \text{or} \quad J_\nu(y_{\nu n}) = -\frac{y_{\nu n}}{\lambda} \frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}}.$$

Substituting these relations allows equation (12) to be simplified in terms of either the function value alone or the derivative value alone

$$\begin{aligned} \left(1 - \frac{\nu^2}{y_{\nu n}^2} \right) J_\nu(y_{\nu n})^2 + \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2 &= \left(1 - \frac{\nu^2}{y_{\nu n}^2} \right) J_\nu(y_{\nu n})^2 + \frac{\lambda^2}{y_{\nu n}^2} J_\nu(y_{\nu n})^2 \\ &= \left(1 + \frac{\lambda^2 - \nu^2}{y_{\nu n}^2} \right) J_\nu(y_{\nu n})^2. \end{aligned}$$

Equivalently, expressing everything in terms of the derivative values

$$\begin{aligned} &= \left(1 - \frac{\nu^2}{y_{\nu n}^2} \right) \frac{y_{\nu n}^2}{\lambda^2} \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2 + \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2 \\ &= \left(1 + \frac{y_{\nu n}^2 - \nu^2}{\lambda^2} \right) \left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2. \end{aligned}$$

Using the classic recurrence identity $\frac{dJ_\nu(x)}{dx} = \frac{1}{2} [J_{\nu-1}(x) - J_{\nu+1}(x)]$ along with the identity $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$, the squared derivative term can be rewritten as

$$\left(\frac{dJ_\nu(y_{\nu n})}{dy_{\nu n}} \right)^2 = \frac{1}{4} [J_{\nu-1}(y_{\nu n}) - J_{\nu+1}(y_{\nu n})]^2$$

$$\begin{aligned}
&= \frac{1}{4} [J_{\nu-1}(y_{\nu n}) + J_{\nu+1}(y_{\nu n})]^2 - J_{\nu-1}(y_{\nu n})J_{\nu+1}(y_{\nu n}) \\
&= \frac{1}{4} \left(\frac{4\nu^2}{y_{\nu n}^2} \right) J_{\nu}(y_{\nu n})^2 - J_{\nu-1}(y_{\nu n})J_{\nu+1}(y_{\nu n}) \\
&= \frac{\nu^2}{y_{\nu n}^2} J_{\nu}(y_{\nu n})^2 - J_{\nu-1}(y_{\nu n})J_{\nu+1}(y_{\nu n}).
\end{aligned}$$

Substituting this result back into the core expression inside equation (12) yields the clean simplification

$$\left(1 - \frac{\nu^2}{y_{\nu n}^2} \right) J_{\nu}(y_{\nu n})^2 + \left(\frac{dJ_{\nu}(y_{\nu n})}{dy_{\nu n}} \right)^2 = J_{\nu}(y_{\nu n})^2 - J_{\nu-1}(y_{\nu n})J_{\nu+1}(y_{\nu n}). \quad (13)$$

In the limit where $\lambda \rightarrow \infty$, the condition simplifies to the standard Dirichlet boundary condition $J_{\nu}(y_{\nu n}) = 0$. Under these conditions, the normalization factor reduces to the classic expression

$$\left(\frac{dJ_{\nu}(y_{\nu n})}{dy_{\nu n}} \right)^2 = [J_{\nu+1}(y_{\nu n})]^2 = [J_{\nu-1}(y_{\nu n})]^2.$$

Throughout this proof, we assumed that the lower boundary values vanish at the origin, specifically that $\rho J'_{\nu}(k\rho)|_{\rho=0} = 0$ and $J_{\nu}(k\rho)|_{\rho=0} = 0$. This is justified by analyzing the asymptotic behavior of the Bessel functions near the origin. Since physical solutions must be regular at $\rho = 0$, the functions behave asymptotically as

$$J_{\nu}(k\rho) = O(\rho^{\alpha}) \quad \text{and} \quad \frac{dJ_{\nu}(k\rho)}{d\rho} = O(\rho^{\alpha-1}) \quad \text{with} \quad \alpha \geq \nu \geq 0.$$

Applying these limits verifies that the required boundary metrics vanish cleanly at $\rho = 0$, confirming the hermiticity of the differential operator and completing the proof.